

SIMATS ENGINEERING CO1 - ASSESSMENT TOOL 1

COURSE NAME: Deep Learning

COURSE CODE: MLA0405

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COURSE OUTCOME COVERED

CO1: Learning Algorithms, Maximum Likelihood Estimation (MLE), Building Machine Learning Algorithms, Neural Networks, Artificial Neurons, Perceptron, Multilayer Perceptron (MLP), Forward Propagation, Backpropagation Algorithm, Gradient Descent, Stochastic Gradient Descent (SGD), Mini-Batch Gradient Descent, Curse of Dimensionality. (BL3)

CO1 Weightage: 15%

Assessment Tool Weightage: 35%

Duration: 1 Hour

Marks: 50

QUESTIONS: 5 Total Marks: 50

Annexure A – Analytical Problem Solving (with Answers)

Question 1

A coin is tossed 10 times and the outcomes are: H H T H H H T H H T

Using Maximum Likelihood Estimation, estimate the probability of getting Heads (p).

Answer:

Step 1 — Problem setup:

Each coin toss can be modelled as a Bernoulli trial with two possible outcomes: Heads (success) with probability p , and Tails (failure) with probability $(1 - p)$. Since the 10 tosses are independent and identically distributed, the overall experiment follows a Binomial distribution with parameters $n = 10$ and unknown probability p .

Step 2 — Count the observations:

From the sequence H H T H H H T H H T, counting each symbol gives: Number of Heads (k) = 7, Number of Tails = 3, Total number of tosses (n) = 10.

Step 3 — Write the likelihood function:

For a Binomial/Bernoulli process, the likelihood of observing k heads in n independent tosses is:

$$L(p) = C(n, k) \cdot p^k \cdot (1 - p)^{(n - k)}$$

Here $C(n, k)$ is a constant (does not depend on p), so maximizing $L(p)$ is equivalent to maximizing $p^k \cdot (1 - p)^{(n - k)}$.

Step 4 — Take the log-likelihood:

It is easier to maximize the log of the likelihood: $\ln L(p) = k \cdot \ln(p) + (n - k) \cdot \ln(1 - p)$

Step 5 — Differentiate and set to zero:

$$d/dp [\ln L(p)] = k/p - (n - k)/(1 - p) = 0$$

Solving this equation gives the general MLE formula: $\hat{p} = k / n$

Step 6 — Substitute the values:

$$\hat{p} = k / n = 7 / 10 = 0.7$$

Conclusion:

The Maximum Likelihood Estimate of the probability of getting Heads is $\hat{p} = 0.7$, i.e. 70%. This means that, based purely on the observed sample, the coin is more likely to land Heads than Tails, and this value is the single estimate of p that makes the observed sequence of outcomes most probable.

Using MLE, the probability of getting Heads is estimated as:

$$\hat{p} = \frac{\text{Number of Heads}}{\text{Total tosses}}$$

Given outcomes:

H H T H H H T H H T

Number of Heads = 7

Total tosses = 10

$$\hat{p} = \frac{7}{10} = 0.7$$

So, the Maximum Likelihood Estimate of getting Heads is:

$$\boxed{p = 0.7}$$

Therefore, the estimated probability of Heads is **70%**.

Question 2

Out of 50 students, 42 passed an exam.

Assuming a Bernoulli distribution, find the MLE of the probability that a student passes.

Answer:

Step 1 — Problem setup:

Let each student's result be a Bernoulli random variable: $X_i = 1$ if the student passes (probability p), and $X_i = 0$ if the student fails (probability $1 - p$). There are $n = 50$ independent students, and we want the MLE of the pass probability p .

Step 2 — Identify the data:

Number of students who passed (successes) = 42. Total number of students (trials) = 50.

Step 3 — Likelihood function:

For n independent Bernoulli trials with k successes, the likelihood is:

$$L(p) = p^k \cdot (1 - p)^{(n - k)}$$

Step 4 — Log-likelihood and differentiation:

$$\ln L(p) = k \cdot \ln(p) + (n - k) \cdot \ln(1 - p)$$

Setting the derivative with respect to p equal to zero and solving gives the standard result:

$$\hat{p} = k / n$$

Step 5 — Substitute values:

$$\hat{p} = 42 / 50 = 0.84$$

For a Bernoulli distribution, the MLE of probability of success is:

$$\hat{p} = \frac{\text{Number of successes}}{\text{Total observations}}$$

Here:

$$\hat{p} = \frac{42}{50} = 0.84$$

So, the Maximum Likelihood Estimate of the probability that a student passes is:

$$\boxed{0.84}$$

Therefore, the estimated pass probability is **84%**.

Conclusion:

The MLE of the probability that a student passes the exam is $\hat{p} = 0.84$, i.e. 84%. This is simply the observed pass rate in the sample, which under the MLE framework is also the value of p that makes the observed data (42 passes out of 50) the most probable outcome.

Question 3

A factory produces light bulbs. In a sample of 100 bulbs, 8 are defective.
Find the MLE for the probability that a bulb is defective.

Answer:

Step 1 — Problem setup:

Each bulb can be treated as a Bernoulli trial: defective (success) with probability p , or non-defective (failure) with probability $(1 - p)$. The sample consists of $n = 100$ independently inspected bulbs.

Step 2 — Identify the data:

Number of defective bulbs (k) = 8. Total sample size (n) = 100.

Step 3 — Likelihood and log-likelihood:

$$L(p) = p^k \cdot (1 - p)^{(n - k)}$$

$$\ln L(p) = k \cdot \ln(p) + (n - k) \cdot \ln(1 - p)$$

Step 4 — Maximize:

Differentiating $\ln L(p)$ with respect to p and setting it to zero gives the general MLE result for a Bernoulli/Binomial model: $\hat{p} = k / n$

Step 5 — Substitute values:

$$\hat{p} = 8 / 100 = 0.08$$

For a Bernoulli distribution, the MLE of probability of defect is:

$$\hat{p} = \frac{\text{Number of defective bulbs}}{\text{Total bulbs}}$$

Here:

$$\hat{p} = \frac{8}{100} = 0.08$$

So, the Maximum Likelihood Estimate for the probability that a bulb is defective is:

$$\boxed{0.08}$$

Therefore, the estimated probability of a defective bulb is **8%**.

Conclusion:

The MLE for the probability that a bulb is defective is $\hat{p} = 0.08$, i.e. 8%. In practical engineering terms, this suggests that, based on the sample, roughly 8 out of every 100 bulbs produced by the factory are expected to be defective, which could be used as an estimate of the defect rate for quality control purposes.

Question 4

A biased die is rolled 60 times. The number 6 appears 18 times.
Estimate the probability of rolling a 6 using MLE.

Answer:

Step 1 — Problem setup:

Since we are only interested in whether a 6 appears or not, each roll can be treated as a Bernoulli trial: success = rolling a 6 (probability p), failure = rolling any other number (probability $1 - p$). There are $n = 60$ independent rolls.

Step 2 — Identify the data:

Number of times a 6 appeared (k) = 18. Total number of rolls (n) = 60.

Step 3 — Likelihood and log-likelihood:

$$L(p) = p^k \cdot (1 - p)^{(n - k)}$$

$$\ln L(p) = k \cdot \ln(p) + (n - k) \cdot \ln(1 - p)$$

Step 4 — Maximize:

Differentiating with respect to p and equating to zero gives the same general MLE formula as before: $\hat{p} = k / n$

Step 5 — Substitute values:

$$\hat{p} = 18 / 60 = 0.3$$

Using MLE:

$$\hat{p} = \frac{\text{Number of times 6 appears}}{\text{Total rolls}}$$

Here:

$$\hat{p} = \frac{18}{60} = 0.3$$

So, the Maximum Likelihood Estimate of rolling a 6 is:

$$\boxed{0.3}$$

Therefore, the estimated probability of rolling a 6 is **30%**.

Conclusion:

The MLE of the probability of rolling a 6 on this biased die is $\hat{p} = 0.3$, i.e. 30%. Since a fair die would give a theoretical probability of $1/6 \approx 0.167$ for any face, the estimated 0.3 confirms that this die is indeed biased toward showing a 6 more often than expected.

Question 5

Given: Input (x) = 4, Weight (w) = 2, Actual Output = 10.

Prediction: $y = w \times x$

(a) Find the predicted output.

(b) Calculate the error.

(c) If the gradient is 4 and learning rate is 0.01, find the updated weight.

Answer:

Part (a) — Predicted output:

The prediction of a single artificial neuron (without bias/activation) is given by the linear equation:

$$y = w \times x$$

Substituting the given values: $y = 2 \times 4 = 8$

Therefore, the predicted output is $y = 8$.

$$[x = 4, \quad w = 2, \quad \text{Actual Output} = 10]$$

(a) Predicted output

$$[y = w \times x] [y = 2 \times 4 = 8]$$

So, the predicted output is:

$$\boxed{8}$$

Part (b) — Error calculation:

The error represents how far the model's prediction is from the actual (target) output. Using the simple difference form of error:

$$\text{Error} = \text{Actual output} - \text{Predicted output} = 10 - 8 = 2$$

This can also be expressed as the Squared Error, which is commonly used as the loss function in gradient-based learning because it is always non-negative and penalizes larger errors more heavily:

$$\text{Squared Error} = (\text{Actual} - \text{Predicted})^2 = (10 - 8)^2 = 2^2 = 4$$

So the error is 2 (absolute/simple error), or equivalently 4 (squared error).

(b) Error

$$[\text{Error} = \text{Actual Output} - \text{Predicted Output}] [\text{Error} = 10 - 8 = 2]$$

So, the error is:

$$\boxed{2}$$

Part (c) - Updated weight using Gradient Descent:

In gradient descent, the weight is updated in the direction that reduces the loss, using the rule:

$$w_{\text{new}} = w_{\text{old}} - (\text{learning rate} \times \text{gradient})$$

(c) Updated weight

Using gradient descent:

$$w_{\text{new}} = w - \eta \times \text{gradient} \quad w_{\text{new}} = 2 - (0.01 \times 4) \quad w_{\text{new}} = 2 - 0.04 = 1.96$$

So, the updated weight is:

$$\boxed{1.96}$$

Here, $w_{\text{old}} = 2$, learning rate (η) = 0.01, and gradient ($\partial \text{Loss} / \partial w$) = 4.

Substituting these values:

$$w_{\text{new}} = 2 - (0.01 \times 4)$$

$$w_{\text{new}} = 2 - 0.04$$

$$w_{\text{new}} = 1.96$$

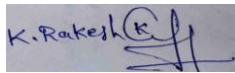
Conclusion:

After one step of gradient descent, the weight decreases slightly from 2 to 1.96. This small step in the direction opposite to the gradient is expected to reduce the prediction error on the next forward pass, which is the fundamental idea behind training neural networks iteratively.

Code of Conduct:

I KOTHAKOTA RAKESH / 192525075 (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.



Signature of the student

Name of the student : RAKESH

RUBRICS FOR CALCULATION

Criteria	Weightage	Level 4 – Exemplary (5 Marks)	Level 3 – Proficient (4 Marks)	Level 2 – Developing (2–3 Marks)	Level 1 – Beginning (0–1 Mark)
Problem Understanding	20%	Clearly interprets problem, identifies all parameters and requirements accurately	Understands problem with minor gaps	Partial understanding; misses key elements	Misinterprets or unable to understand problem
Method Selection	20%	Selects most appropriate method/formula with strong justification	Selects correct method with minor errors	Inappropriate or partially correct method	Unable to select method
Solution Process	25%	Logical, systematic steps with correct approach throughout	Mostly correct steps with minor mistakes	Disorganized steps; multiple errors	No clear process
Accuracy of Calculation	20%	All calculations accurate; correct final answer	Minor calculation errors	Frequent errors; incorrect final answer	No valid calculations
Interpretation of Results	15%	Clearly interprets results with units and engineering relevance	Basic interpretation with minor omissions	Limited or unclear interpretation	No interpretation